



Business Process Map Administrator Manual

Consoles, operations, and governance for system administrators. Everything in the User Manual applies to you too.

Company process-map system · As of 2026-09-12 · English edition

Navigate with   or click the screen edges ·  to restart

What's inside

01 The Sysadmin Role

03 Content Management

05 Framework Management

07 Approvals · Groups · Trash

09 Configuration Reference

02 Admin Console Map

04 Directory & Org

06 Database & Backups

08 Dashboard

01

The Sysadmin Role

How sysadmin is granted and what it unlocks.

How it's granted

- Not stored in the DB — controlled by the `BPM_SYSADMINS` env var (comma-separated login IDs), read at startup.
- With auth off (dev default), effectively everyone is sysadmin.
- With `DEV_ENFORCE_PERMISSIONS=true` or real auth, only listed IDs are sysadmin.
- In ldap auth mode, the Local Accounts Sysadmin toggle can grant it too (effective only there).
- The **Settings** → **Directory** → **Employees** table tags them Sysadmin (tooltip *managed by environment*) — **env-listed users** can't be turned off from any screen.

What sysadmin unlocks

- Treated as **Owner on every map** — manage collaborators, approvers, visibility, versions, deletion.
- **Force checkout** — the only role that can take another user's active edit lock.
- Every **admin-only console** under Settings.
- Moderation — group approvals, feedback replies/status, the global trash, manual publishing.
- **Withdraw is the exception** — Pending/Approved versions only by the submitter (Rejected also by owner/sysadmin).

02

Admin Console Map

Every console on the Settings rail, at a glance.

The Settings rail

- **Content** — Notices · Manual · **Knowledge base** · AI chat · **AI prompts**.
- **Directory** — Employees · Departments · **Catalogs** · **Orphaned refs** · Local Accounts (Idap mode only).
- **Framework** — Categories & import (**Manage / Status** views, linkage admins). Delegated admins see only their subtree.
- **Database** — Tables · Batch jobs (**Back up now**, backup downloads). **Approvals** — Approval Queue.
- Groups · Analytics (Dashboard) · Trash (Scheduled deletion).

Business Process Map

Maps Notices Inbox 2

Feedback System Admin 한 EN

Settings

CONTENT

Notices

Manual

Knowledge base

AI chat

AI prompts

DIRECTORY

Employees

Departments

FRAMEWORK

Categories & import

DATABASE

Tables

Batch jobs

APPROVALS

Approval Queue 2

GROUPS

User Groups

ANALYTICS

Dashboard

TRASH

Scheduled deletion

Notices

2 total · 2 live

Export CSV + New notice

Status	Title	Importance	Period
Live	Scheduled maintenance — Sep 6 (Sat) 22:00-24:00	General	08-31 ~ Unlimited
Live	BPM 프로세스맵 정식 오픈 안내	Important	08-31 ~ Unlimited

03

Content Management

Notices, the in-app manual, AI knowledge/settings/prompts, and feedback.

Notices

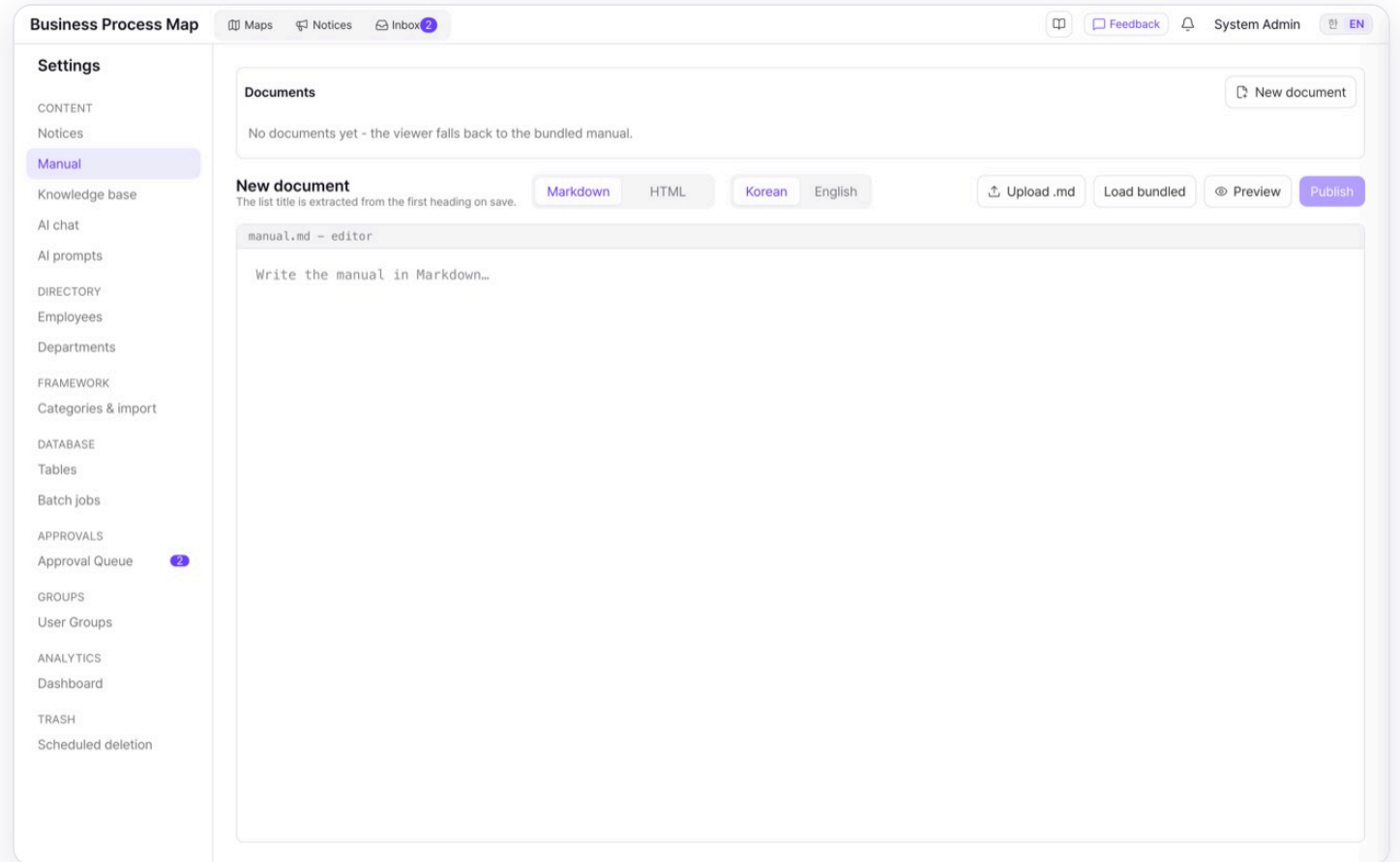
- Title, importance (Important/General), posting period (open-ended allowed), **Markdown body**.
- Check **Notify all users on publish** to push to every inbox.
- Users see only active notices; the admin list shows all, with **CSV export**.
- **Deletion is hard** — permanent, no trash (already-sent bell notifications remain).

The screenshot shows the 'Business Process Map' interface. On the left is a sidebar with a 'Settings' section containing a tree view: CONTENT (Notices, Manual, Knowledge base, AI chat, AI prompts), DIRECTORY (Employees, Departments), FRAMEWORK (Categories & import), DATABASE (Tables, Batch jobs), APPROVALS (Approval Queue), GROUPS (User Groups), ANALYTICS (Dashboard), and TRASH (Scheduled deletion). The 'Notices' item under 'CONTENT' is selected. The main area is titled 'Notices' with a sub-header '2 total · 2 live'. It includes an 'Export CSV' button and a '+ New notice' button. Below is a table with columns: Status, Title, Importance, and Period. Two notices are listed, both with a 'Live' status.

Status	Title	Importance	Period
Live	Scheduled maintenance — Sep 6 (Sat) 22:00-24:00	General	08-31 ~ Unlimited
Live	BPM 프로세스맵 정식 오픈 안내	Important	08-31 ~ Unlimited

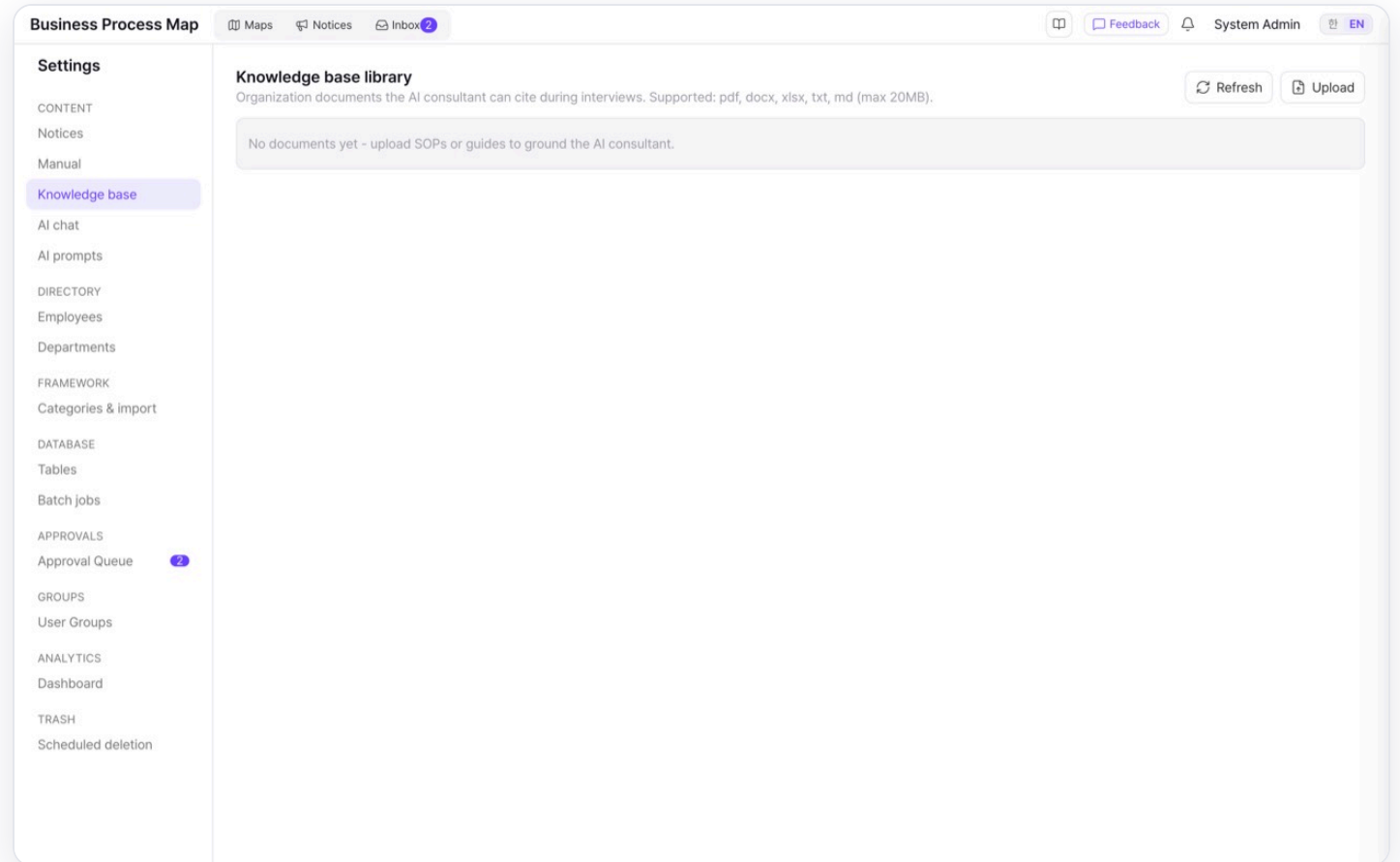
Publishing the manual

- `/manual` holds several documents — each with a format (markdown / sanitized HTML) and a language (KO/EN).
- Write inline, upload an .md, or load the bundled copy; preview, then publish for an immediate switch.
- List titles come from the first heading — upload KO and EN in the same order so the language toggle opens the matching document.
- With no publish history the bundled fallback is served and flagged with a bundled badge.
- Click a row to edit, **New document** to add, the row's trash to delete. The viewer builds its TOC from `##` / `###` headings — images and raw HTML don't render.



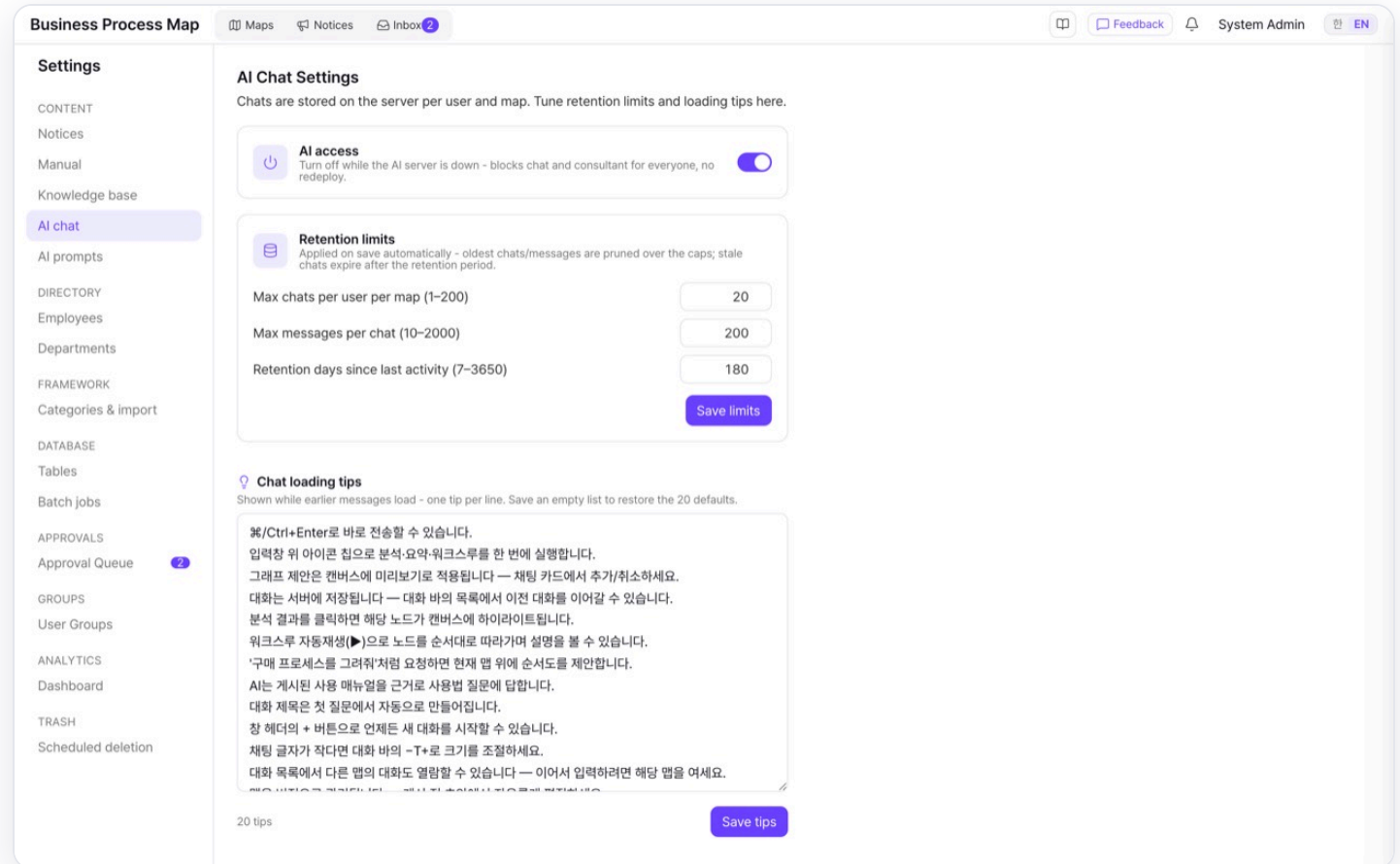
Knowledge base

- A library of org documents the AI consultant cites in interviews — upload SOPs and guides.
- Supports pdf, docx, xlsx, txt, md — up to 20 MB per file. Add with Upload, reload with Refresh.
- Works only when both `AI_ENABLED` and the embedding server `EMBED_URL` (📍) are set.



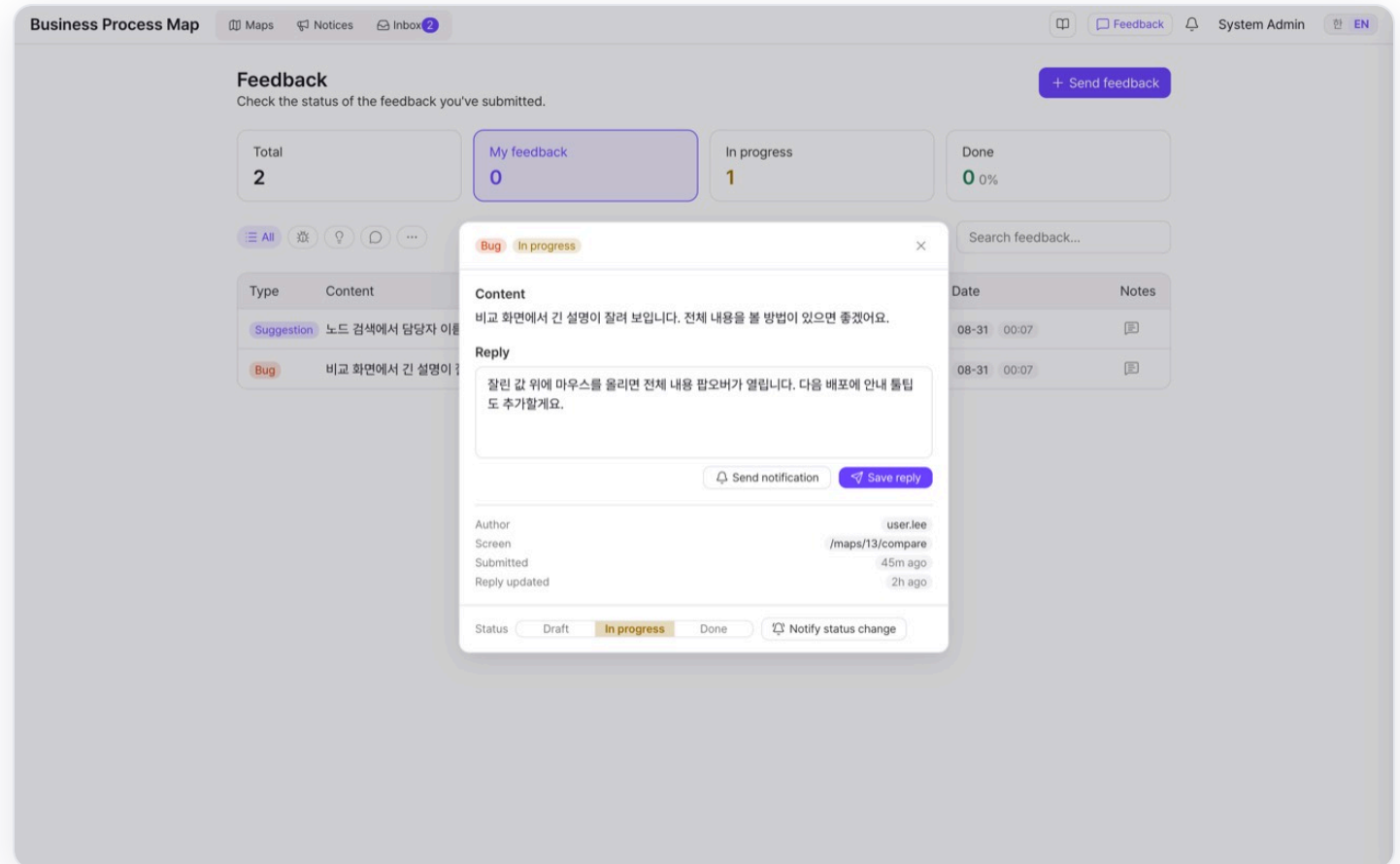
AI chat settings · AI prompts

- The **AI access** switch at the top pauses every AI feature (chat, interview) without a redeploy — effective availability = `AI_ENABLED` AND the switch.
- Chats are always stored server-side (scoped to user × map, owner-visible) — not a toggle.
- Retention caps — 20 chats/map, 200 messages/chat, 180 days (tunable, applies immediately). Loading tips: one per line; clearing restores the 20 defaults.
- AI prompts — override the built-ins: saves apply instantly; clearing falls back to the shipped prompt.
- While overridden, prompt improvements shipped in code are not picked up — re-check periodically.



Feedback administration

- Only sysadmin changes status and replies — **done** locks further replies.
- Notifications are deliberate — **Send notification** (reply, resendable) and **Notify status change** (once per item).
- The "Notification sent <time>" meta row records what went out.
- **Notes** — anyone writes; edits keep history; delete = archive. Permanent purge only via the `feedback_notes` table view.



04

Directory & Org

HR-webhook sync, the departments basis, heads, and consultant accounts.

Employees & sync

- The source is the n8n HR webhook — a full sync refreshes employees plus the departments table (AD only enriches titles).
- The Employees tab's **Sync all from AD** button (label kept) — spinner while running, then a scanned/upserted/deactivated/deleted/skipped summary. One per 5 min.
- Auto sync every `HR_SYNC_INTERVAL_HOURS` (default 24 h, 0 = off), plus per-user daily sync on login.
- Safeguards — the sync-preview dry-run API and the deletion cap `HR_SYNC_DELETE_CAP_PCT` (default 20%).
- Departures aren't deleted — marked active=false and excluded from pickers.

Business Process Map Maps Notices Inbox 2

Settings

CONTENT

Notices

Manual

Knowledge base

AI chat

AI prompts

DIRECTORY

Employees

Departments

FRAMEWORK

Categories & import

DATABASE

Tables

Batch jobs

APPROVALS

Approval Queue 2

GROUPS

User Groups

ANALYTICS

Dashboard

TRASH

Scheduled deletion

Employees Export CSV Sync all from AD

Exposed positions

Choose which EDW titles are shown as a person's position across the app.

☒ 그룹장 ☒ 파트장 ☒ 팀장 ☒ 센터장

Save

loginid	name	korean name	korean dept	title	department	role	status	source
admin.sys	System Admin	Sysadmin		Manager	System Team	admin	Active	local
bora.choi	Bora Choi			Lead	Support Team	user	Active	local
bora.hong	Bora Hong			Lead	Brand Part 1	user	Active	local
bora.hwang	Bora Hwang			Manager	Planning Part 1	user	Active	local
bora.jang	Bora Jang			Manager	Brand Part 2	user	Active	local
bora.jung	Bora Jung			Manager	Growth Team	user	Active	local
bora.jung2	Bora Jung			Senior	Brand Part 3	user	Active	local
bora.kwon	Bora Kwon			Lead	Brand Part 2	user	Active	local
bora.lee	Bora Lee			Senior	Planning Part 1	user	Active	local
bora.lee2	Bora Lee			Senior	Facilities Office	user	Active	local
bora.lee3	Bora Lee			Associate	Planning Part 1	user	Active	local
bora.oh	Bora Oh			Associate	Insights Team	user	Active	local
bora.oh2	Bora Oh			Staff	Planning Part 2	user	Active	local
bora.park	Bora Park			Senior	Logistics Part 1	user	Active	local
bora.seo	Bora Seo			Senior	Analytics Part 1	user	Active	local

Departments & heads

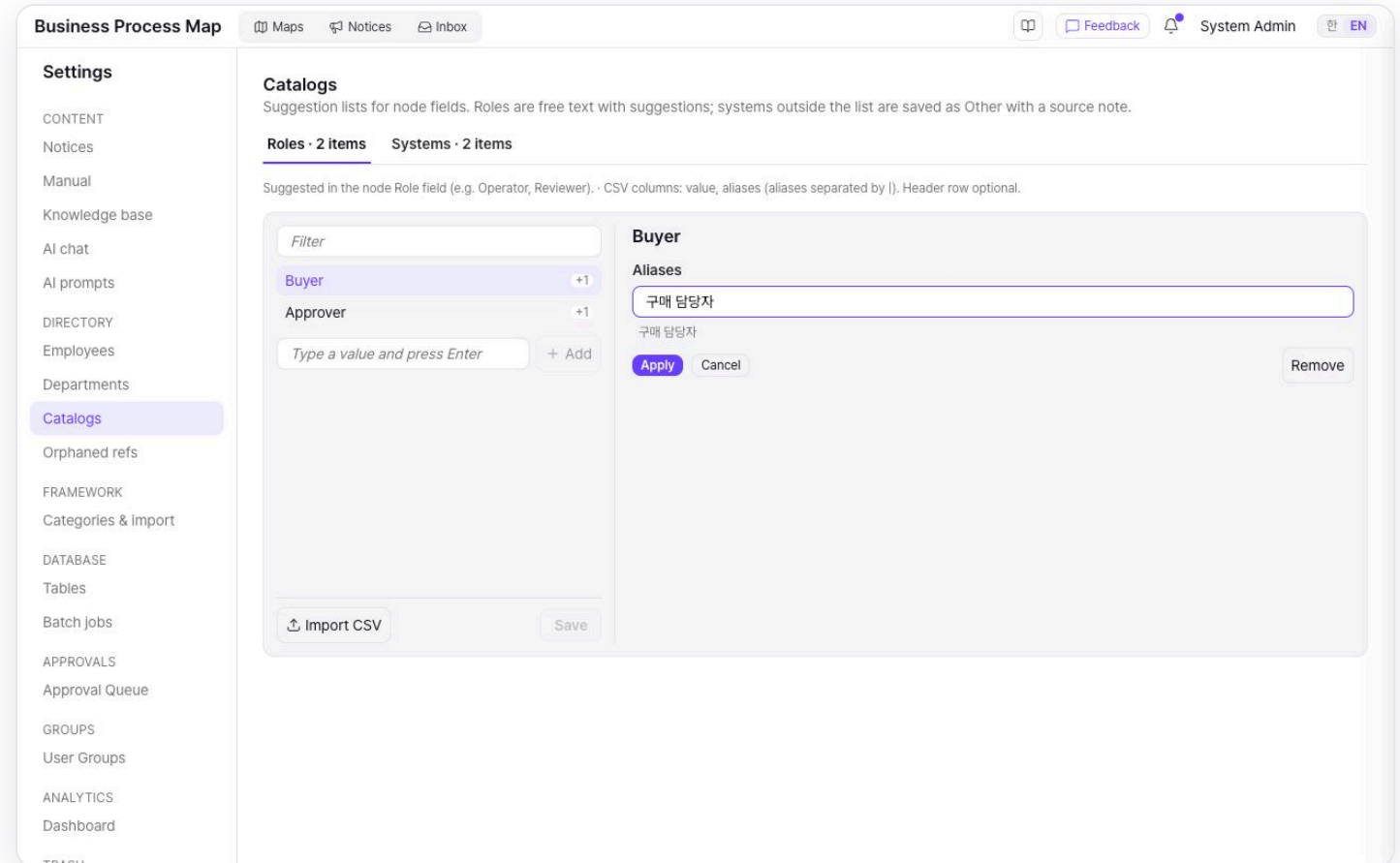
- The departments table is the org basis — the single source for the org chart and path resolution.
- Heads are determined from EDW position (FRNM) data, tagged Manager on the chain.
- Exposed positions — a card on the **Employees tab** picks which collected EDW titles show in the app (defaults: 그룹장·파트장·팀장·센터장); an app setting, not env.
- Stale department and departed-person references are cleaned up on the **Orphaned refs** tab (two slides on) — it replaces the old Missing departments remap.
- Both tables support CSV export (BOM, formula-injection guard).

The screenshot shows the 'Business Process Map' application interface. On the left is a sidebar menu with categories: CONTENT (Notices, Manual, Knowledge base, AI chat, AI prompts), DIRECTORY (Employees, Departments), FRAMEWORK (Categories & import), DATABASE (Tables, Batch jobs), APPROVALS (Approval Queue), GROUPS (User Groups), ANALYTICS (Dashboard), and TRASH (Scheduled deletion). The 'Departments' item is highlighted. The main area displays a table with columns 'Department', 'korean dept', and 'Members'. The table lists various departments and their member counts, with some rows expanded to show sub-departments.

Department	korean dept	Members
▼ Growth Center		0
▼ Marketing Office		0
▼ Brand Team		0
Brand Part 1		31
Brand Part 2		31
Brand Part 3		31
Growth Team		31
▼ Strategy Office		0
▼ Analytics Team		0
Analytics Part 1		31
Insights Team		31
▼ Planning Team		0

Catalogs (Roles & Systems)

- Autocomplete lists for node **Role** and **System** — **Role | System** tabs, the item list on the left (filter, add), the selected item's **alias** editor on the right.
- Typing an alias saves the canonical spelling (`sap` → SAP ERP). **One alias belongs to one item** — a colliding alias is dropped on save with a notice.
- Import a two-column `value,aliases` CSV (aliases split by `|`, header optional) with an added/duplicate preview; the System tab offers **values in use** to promote.
- **Other** in the System list is reserved (cannot be deleted, always on top). Everyone reads the lists; **editing and saving is sysadmin-only**.
- The editor, CSV import, AI assistant, consultant interview and interview JSON import all normalize against these lists — unknown systems become Other + source note, unknown roles stay free text.



Orphaned reference audit (Orphaned refs)

- **Rescan** scans 12 places on demand — node departments/assignees, SP designations, owners/collaborators/approvers, owning departments, user groups, category admins.
- Results group **by value** (vanished department path, departed person); expanding lists each reference (map, version, node, role) with a checkbox.
- **Bulk replace** checked lines with a current department or person, or **remove** them. Node fields are display-only — fix them in the editor draft.
- Send owners a **Stale refs fix request** (one per owner, map list attached); when the owner has left, assign a new one (**Owner assigned**).
- Users see a **Stale refs** badge, the home **Issues** filter and warning-colored node icons. Replaces the old Missing departments tab.

The screenshot shows the 'Business Process Map' interface with a sidebar on the left and a main content area on the right. The sidebar includes sections like CONTENT, DIRECTORY, FRAMEWORK, DATABASE, APPROVALS, GROUPS, ANALYTICS, and TRASH. The 'Orphaned refs' option is highlighted in the DIRECTORY section. The main content area displays a message: 'References to departments or people that no longer exist in the current org. Check the lines to reassign in bulk; node fields must be fixed by the map owner in a draft.' Below this, there are two sections: 'Missing departments' (2 items) and 'Departed users' (6 items). Each item is a row with a red warning icon, a text description, and a 'login' button. The 'Missing departments' section shows two identical entries for 'Quality Center/QC Department/QC Support Team/QC Sample Management Group'. The 'Departed users' section shows six entries for various users like 'Admin Sys', 'cheolsu.kim', 'slot.l5admin', 'slot.owner', 'user.choi', and 'younghee.lee'.

Section	Item	Action	Role
Missing departments (2)	Quality Center/QC Department/QC Support Team/QC Sample Management Group	leaf	Subprocess department 3
	Quality Center/QC Department/QC Support Team/QC Sample Management Group	path	Owning department 3
Departed users (6)	Admin Sys	name	Node assignee 19
	cheolsu.kim	login	Approver 3
	slot.l5admin	login	Category admin 1
	slot.owner	login	Collaborator 2 · Owner 2
	user.choi	login	Collaborator 2 · Owner 3
	younghee.lee	login	Approver 2

Local accounts (ldap mode)

- Appears only in ldap auth mode — issue ID/password logins for external consultants.
- Sign-in checks local accounts first, then falls back to AD. Throttled to 5 attempts per 5 minutes.
- Per account: reset password, deactivate/reactivate, delete (permanent). Deactivate first when an engagement ends.
- Sysadmin toggle — **local accounts only** (directory accounts only via `BPM_SYSADMINS`), effective in ldap mode; env-granted users can't be turned off here.
- Token lifetime and secret rotation (kills all sessions) live in docs/deploy/deploy.md §2.1.

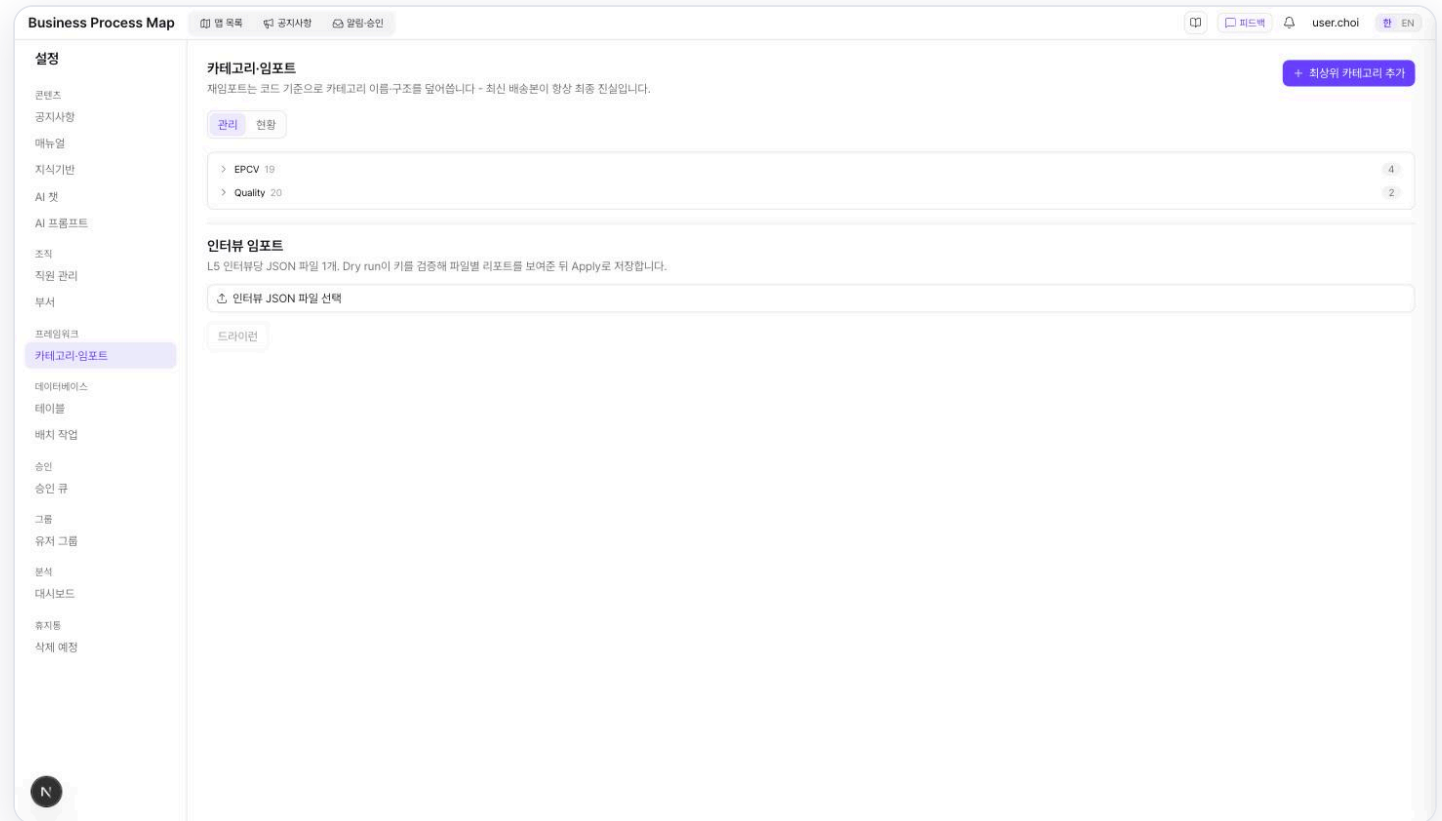
05

Framework Management

Category tree and level delegation, linkage admins, confirmation governance, slot changes, the status board, interview import.

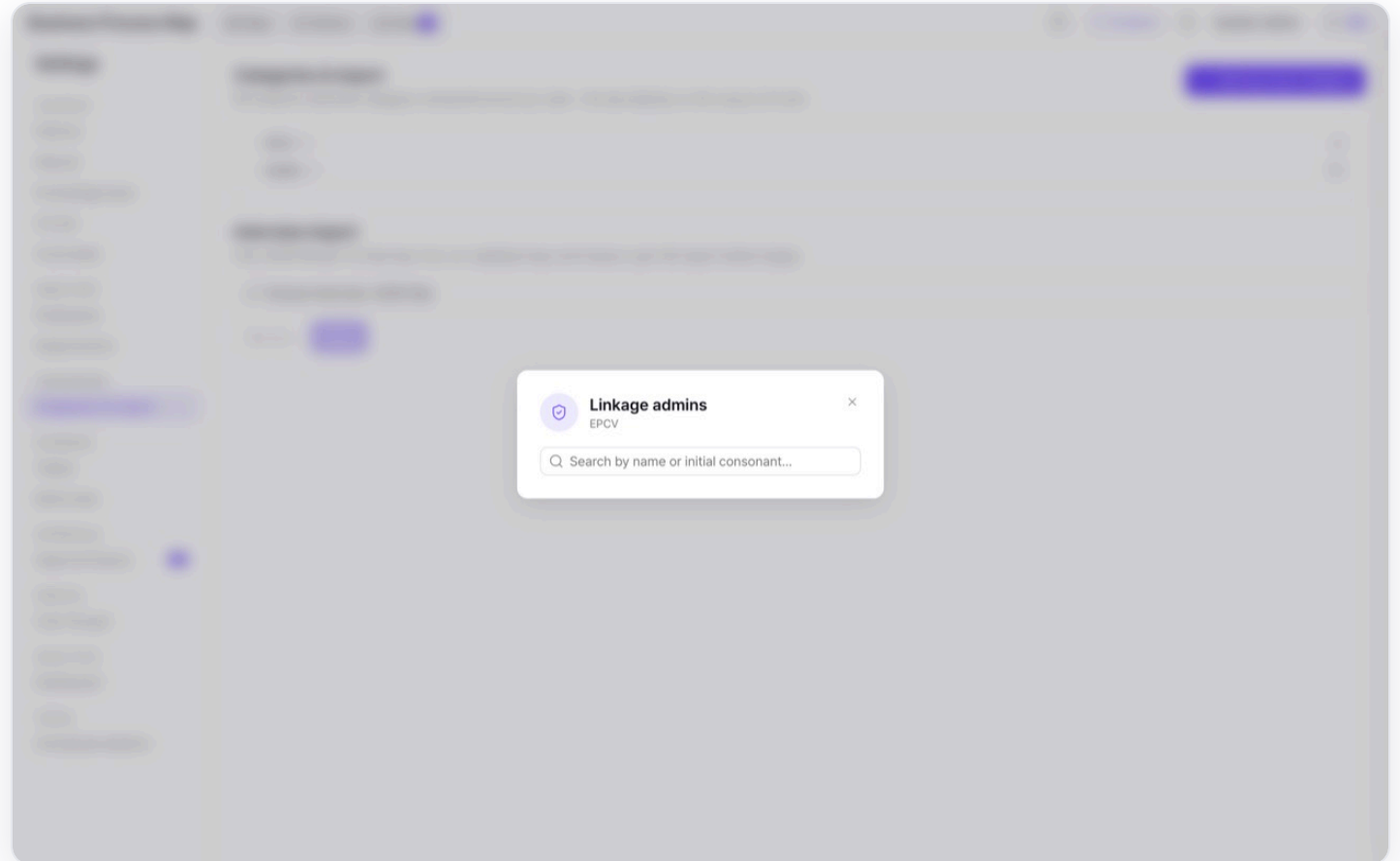
Category management & level delegation

- The top toggle switches between the **Manage** view (tree · admins · import) and the **Status** view (confirmation board).
- Add top/child, rename, move, delete (max 5 levels) — **maps attach to L5 only**. Deletion is refused while the subtree has linked maps.
- Renames sync the linkage-canvas name. **A subtree with a canvas can't be deleted (409)** — clean up the canvas first.
- **Delegation scope** — admins add/rename/reorder under their own category. The delegated category itself can't be moved or deleted (only its children); admins can only be appointed one level down.
- **L5-only admins** edit and confirm the canvas — nothing structural. Top-level categories and interview import stay **sysadmin-only**.



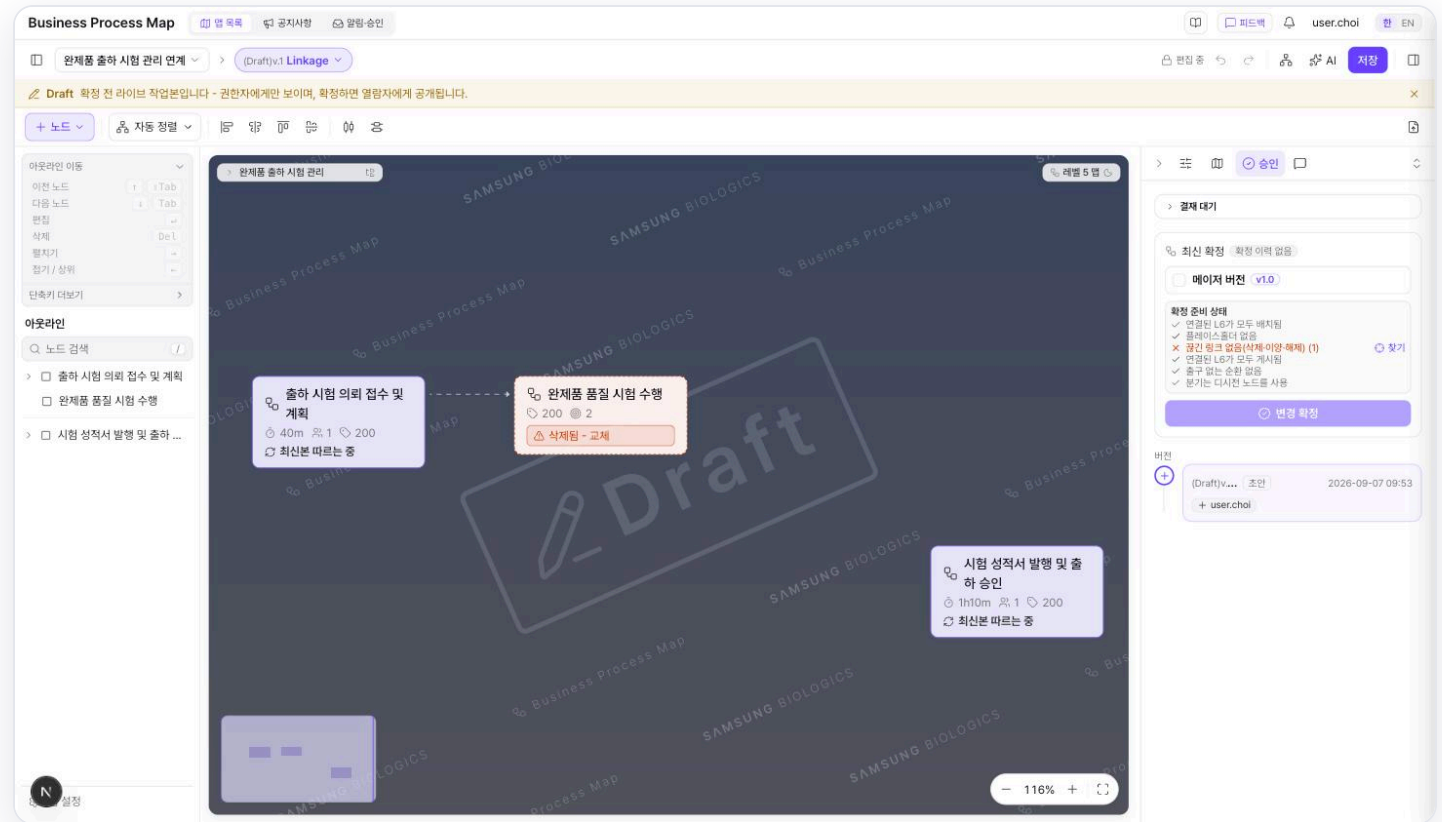
Linkage admins

- The admins button on a category row appoints users/groups. Adds/removes **buffer until you confirm once**; Cancel, Esc, or clicking outside discards.
- Grants **inherit downward** — appoint at L1–L4 and every L5 canvas below becomes editable.
- Tree rows show the **directly appointed admins** inline next to the code (two names, then a **+N** tooltip), dual-named per the language setting.
- Canvases are real maps (`mode=framework`) but never appear in the map list or SP library.
- A major confirm permanently prunes the previous major's **intermediate minors** (X.0 and the last minor are kept, with a preview dialog).



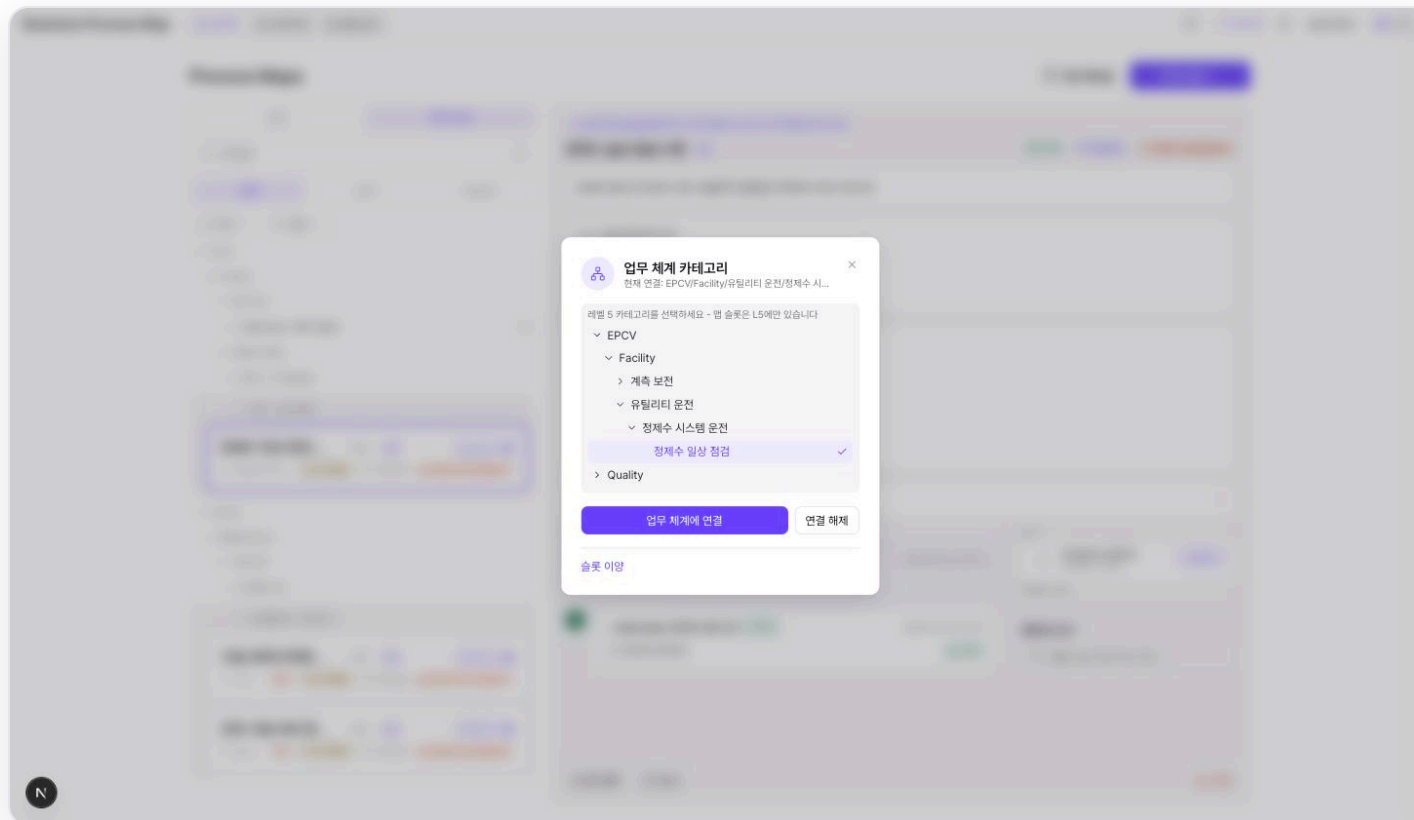
Confirmation governance

- Snapshots are a **confirmed** state apart from publishing — neither they nor the live draft can be deleted; regular map workflows are **blocked server-side**.
- **Six confirm gates** — all L6 placed · no placeholders · no stale links · all L6 published · no exit-less loops · branches use a decision (all-parallel fan-out exempt).
- The Approval tab's **Confirm readiness** checklist is the single source — one failing gate disables Confirm.
- **Who confirms** — the checkout holder, if they are that L5's **direct admin** or a sysadmin. Higher-level admins send a **confirm request** (checkout auto-releases) decided by the direct admin + sysadmin.
- **Draft visibility** — only admins and sysadmins see the live draft; everyone else lands on the latest confirmed snapshot.



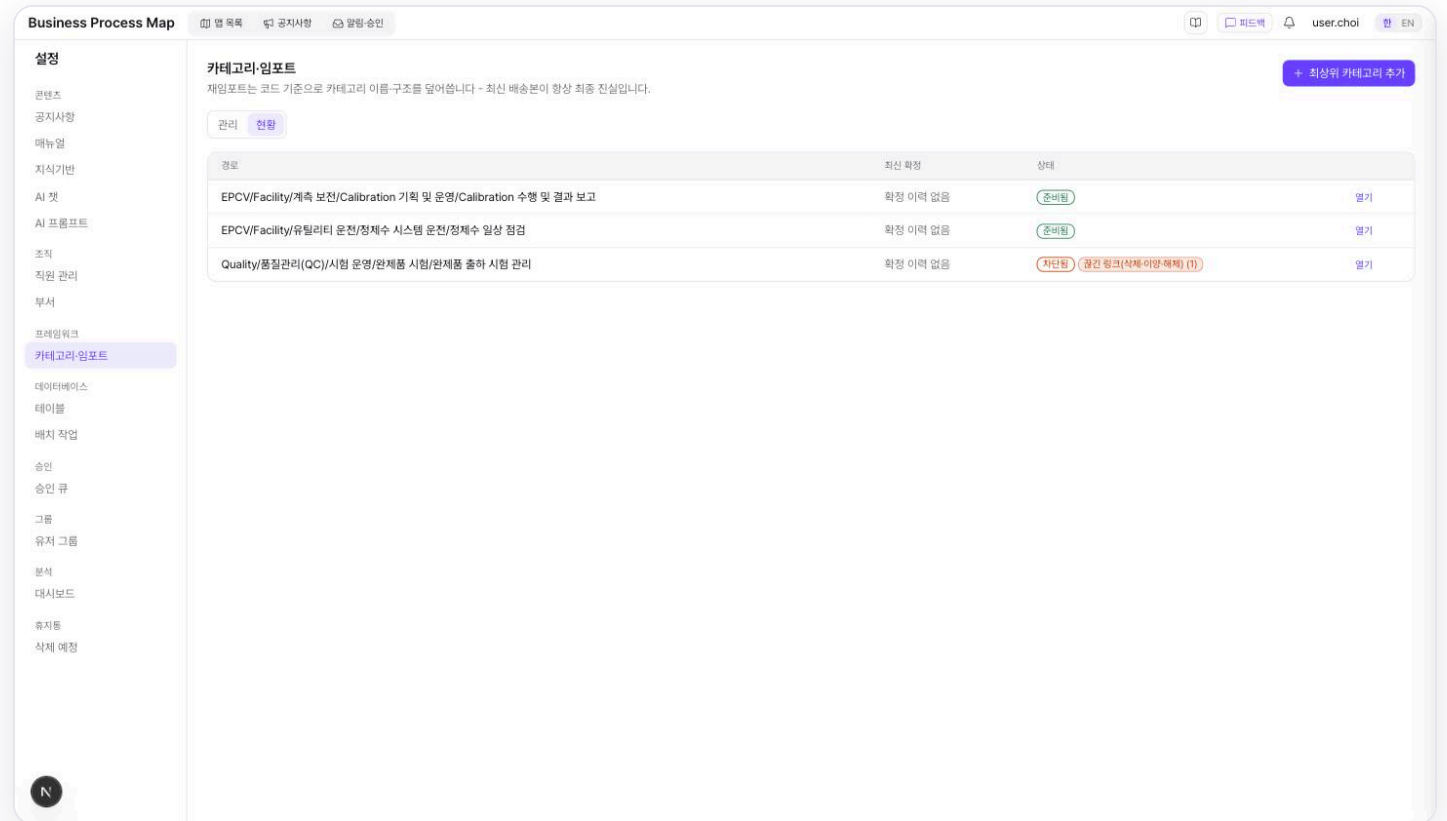
Slot changes

- Which L5 a map belongs to is a **slot** (assign · unassign · move · hand off · delete) — worked from the **Framework pill** on the home card.
- The decider is **that L5's direct admin** (sysadmins too). Higher-category admins can edit the canvas but only **request** a slot change.
- A move between two L5s needs **both L5s' direct admins** (one admin covering both counts once); an L5 without admins falls to sysadmin.
- A requester who is already a direct admin applies it via **Apply now** (toast); otherwise an `fw_slot` request lands in the L5 admins' Inbox, the map's Approval tab and the **Approval Queue** (㉞) — withdrawable by the requester, the current owner or a sysadmin.
- Sysadmins decide from the queue as the **fallback decider** — the map's approvers cannot decide slot requests.



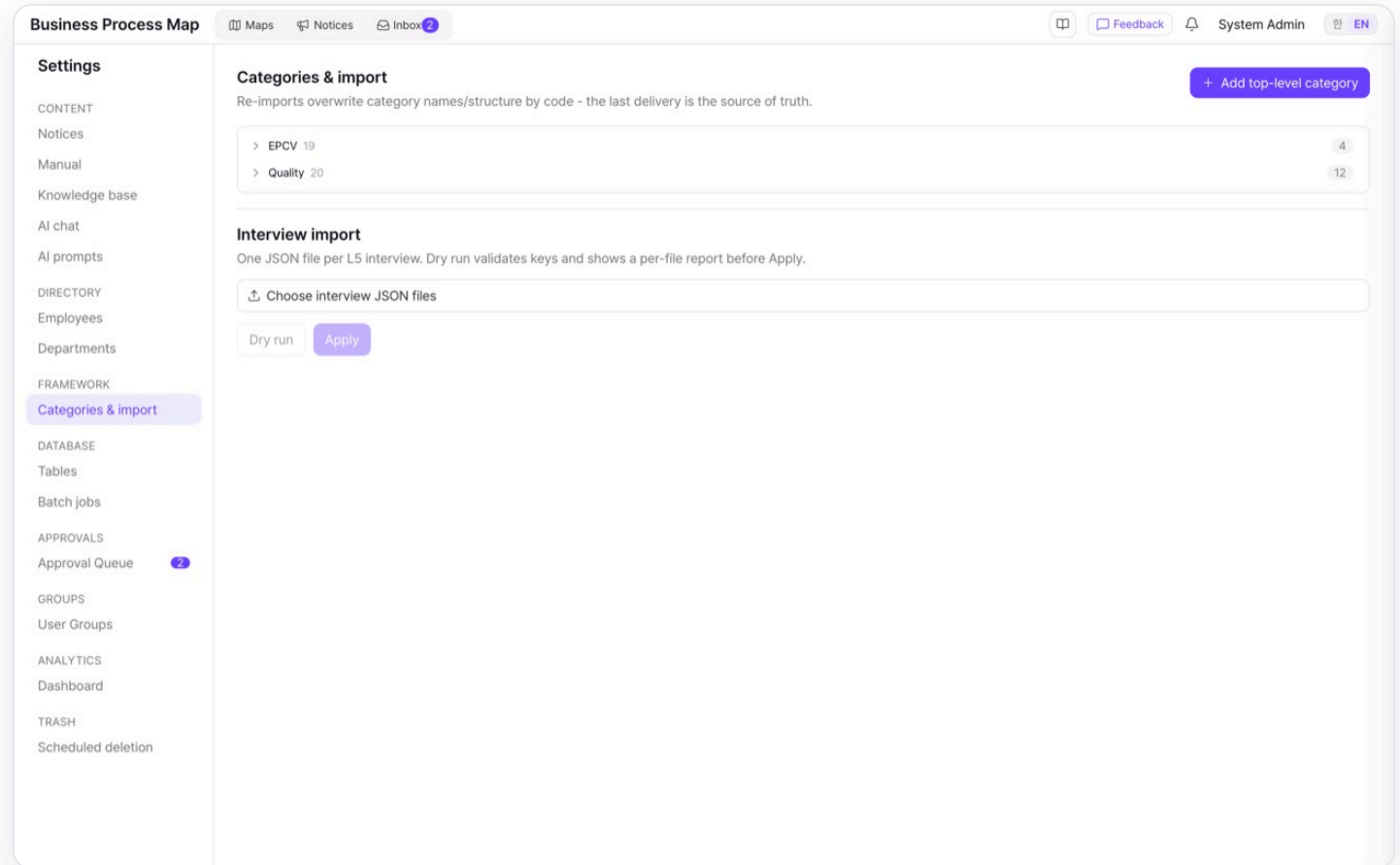
Status board

- The **Status** view lists every L5 in scope — **Path / Latest confirmed / Status** (Ready · Blocked · No canvas) with an **Open** button.
- Blocked canvases carry their failing gates as **negative pills** — Missing L6 · Placeholders · Stale links · Unpublished L6 · Exit-less loop · Direct fan-out.
- The home Framework view's **category summary card** shows the same verdicts under "Subtree confirmation".
- Delegated admins see the status of their own subtree only.



Interview import — format, dry run, governance

- Upload L5 interview JSON files (multi-select) → **Dry run** → **Apply** — sysadmin-only.
- **Format** — interview JSON **0.4** (`relations` flow graph) and **0.5** (`externalTasks`); 0.3 rejected. Error files skip whole; re-runs idempotent.
- **Two-column dry-run report** — left: Summary, Needs review, External L6, Category admins, Governance changes; right: file cards → L5 canvas → maps (row previews).
- **Governance changes (re-import)** — differing owner, department, approvers, imported notes become rows with **Keep** ↔ / **Replace** ⇌; unchanged notes: no row, changed: -/+ diff.
- **Owner assignment** — an empty/unknown owner makes the importing sysadmin interim owner with an **Owner unconfirmed** pill → hand over via Transfer ownership.



Interview import — landing rules, placeholders, notes

- **Department path resolution**, four passes — exact → leading levels dropped → unique suffix → mirror chain. No match registers the delivered path as-is (remap under Settings → Directory → Departments).
- **Landing** — activities (L7) become nodes, flow edges (seq/branch/loop/bypass) become connectors; branch promotes a decision, a self-loop synthesizes a "**Repeat? (auto-generated)**" branch.
- Conditions, times and systems land as map fields with originals kept as **interview source notes**. `ownerRole` lands as the SP designation **Role**; systems are normalized to the catalog (alias → canonical, unknown → **Other + source note**, one report row per map). Annual count and FTE land on both the SP reference and the L5 node.
- Exact output=input matches auto-link as IO links; nodes auto-lay out horizontally; an editable draft is created right after publish (an untouched draft is reused on redelivery).
- **L5 canvas** — built or extended from top-level L6 flows. Redelivery only **adds** nodes and skips a canvas someone else has checked out.
- **External taskId placeholders** — an unknown taskId stays as a placeholder titled with its code; later deliveries or slot assignment resolve it automatically.
- Exceptions, VOC, regulatory basis and open items become map notes; L5 entry/flow/open items become category notes. Settle GMP and conditions on the map's **Conditions & GMP** card (kept across redeliveries).

06

Database & Backups

The table viewer, notification purge, batch status, and daily backups.

The table viewer

- A **read-only** browser with server-side paging, sorting, filtering — it never writes.
- **CSV export** streams all rows with the current sort/filter (BOM, formula guard).
- **Login records** — one row per user per day; the audit view.
- Selecting `notifications` adds a **range purge** — grouped preview, then a hard delete (no undo).
- Notification retention is a fixed most-recent-100-per-person policy.

Business Process Map

Maps Notices Inbox 2

Feedback System Admin EN

Settings

- CONTENT
- Notices
- Manual
- Knowledge base
- AI chat
- AI prompts
- DIRECTORY
- Employees
- Departments
- FRAMEWORK
- Categories & import
- DATABASE**
- Tables
- Batch jobs
- APPROVALS
- Approval Queue 2
- GROUPS
- User Groups
- ANALYTICS
- Dashboard
- TRASH
- Scheduled deletion

Tables

ai_chat_messages 0 ai_chat_sessions 0 ai_prompts 0 ai_usage_events 0 app_settings 0 approval_requests 1 batch_job_runs 3

category_permissions 0 checkout_requests 1 comments 2 dashboard_coverage_depts 0 dashboard_permissions 0 departments 0

edges 206 employees 401 feedback 2 feedback_note_revisions 0 feedback_notes 1 groups 1 interview_attachments 0

interview_checkpoints 0 interview_messages 0 interview_sessions 0 kb_chunks 0 kb_documents 0 local_credentials 0 login_records 3

manual_docs 0 map_approvers 28 map_notes 29 map_permissions 117 map_versions 95 nodes 240 notices 2 **notifications 3**

process_categories 19 process_maps 34 user_group_managers 0 user_group_members 26 user_groups 6 version_approvals 130

version_events 375

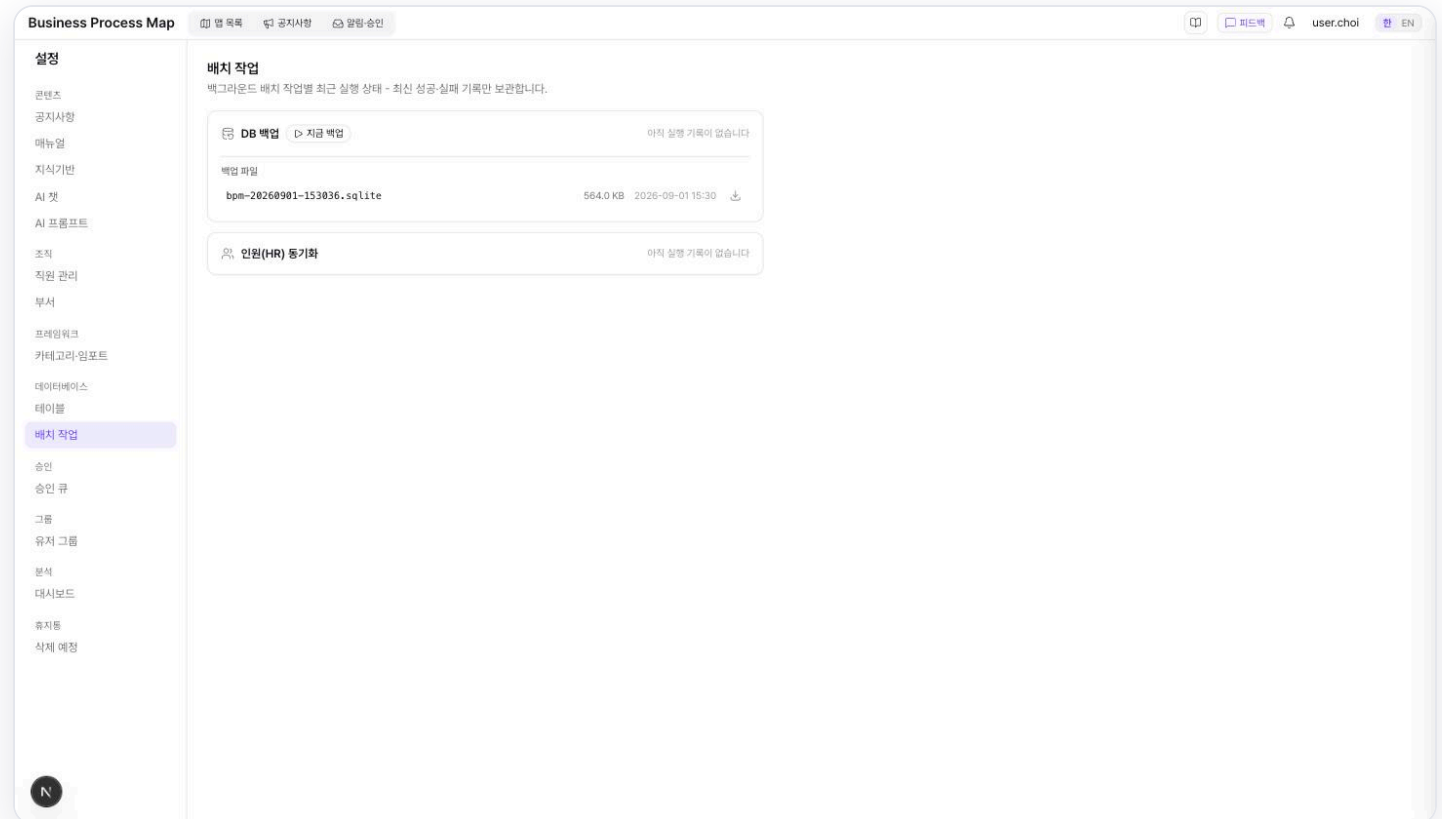
notifications Filter rows... Export CSV 3 rows · 3 shown 연도, 월, 일. - 연도, 월, 일. Delete in range

id	recipient	type	map_id	version_id	message	payload	read	created_at
1	admin.kim	review_requested	14	75	user.lee 님이 'As-Is' 승인을 요청했습니다.		false	2026-08-30T23:59:10.421538
2	admin.kim	review_requested	15	76	user.park 님이 'As-Is' 승인을 요청했습니다.		false	2026-08-30T23:59:10.422816
3	admin.kim	notice	17		새 공지: v2.4 승인 워크플로우 개편 안내가 등록되었습니다.		true	2026-08-30T23:59:10.424745

All rows loaded

Batch job status

- The latest attempts of the DB backup and HR sync, per job — last success and last failure each keep their time and summary.
- A failure newer than the last success is your signal to act.
- **Back up now (on demand)** — a backup outside the daily schedule. On production (postgres) the sidecar picks it up within seconds and runs `pg_dump` ; local sqlite copies instantly.
- **Backup file list & download** — the card lists `${BACKUP_DIR}` and lets you download each file for an off-server copy (sysadmin only).



Daily automatic backups (prod)

- The compose `db-backup` sidecar — daily after **04:00 KST**, plus once on boot if today's dump is missing.
- A dump is kept **only after** `pg_restore --list` verification; outcomes land on the Batch jobs tab.
- Stored at `${BACKUP_DIR}` (default `./backups`); `${BACKUP_RETENTION_DAYS}` (default 14) prunes older dumps.
- Server-disk only for now — **off-server copies recommended**. Recovery runbook: `docs/deploy/backup.md`.

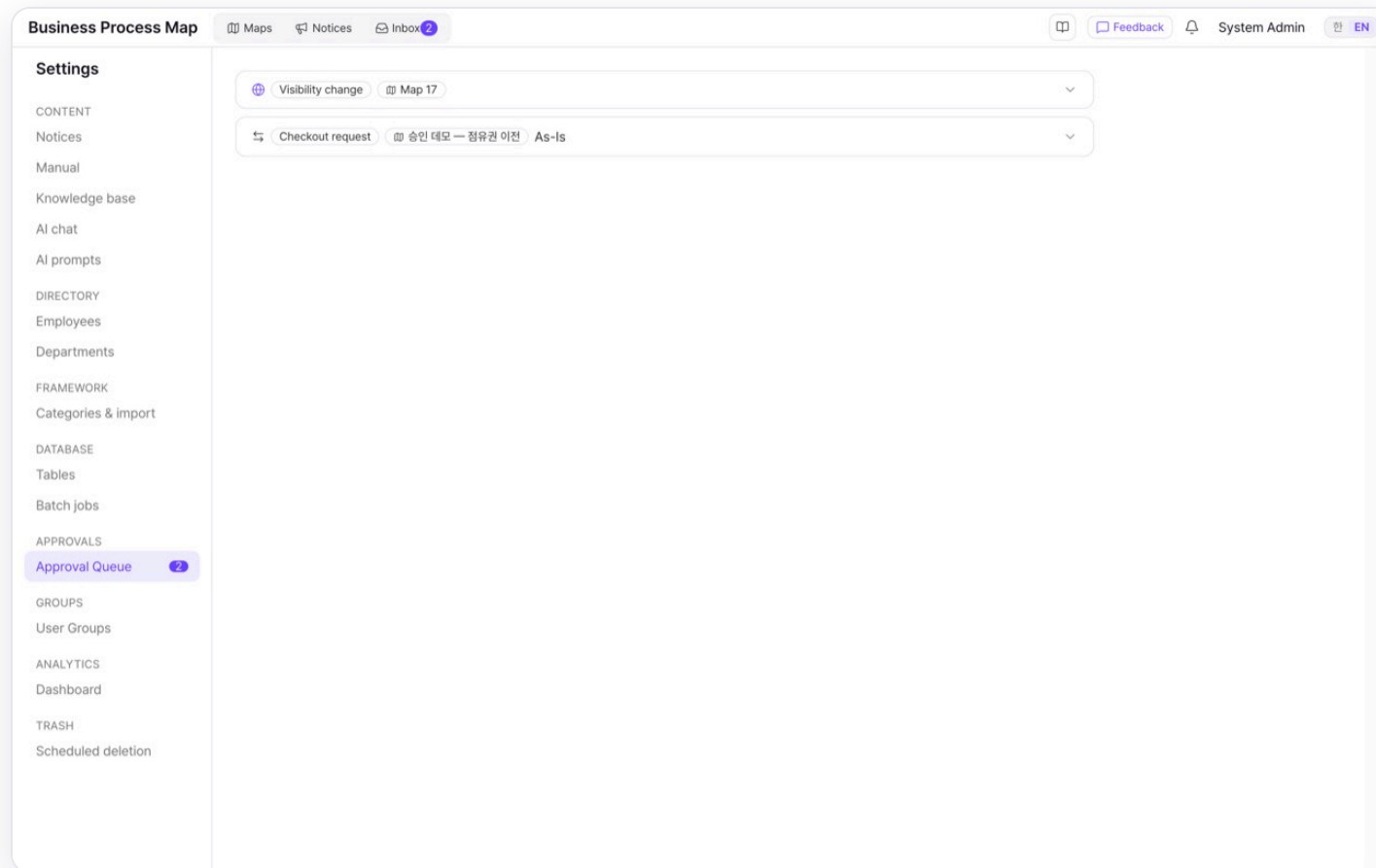
07

Approvals • Groups • Trash

The global queue, group approvals, restore, and instant purge.

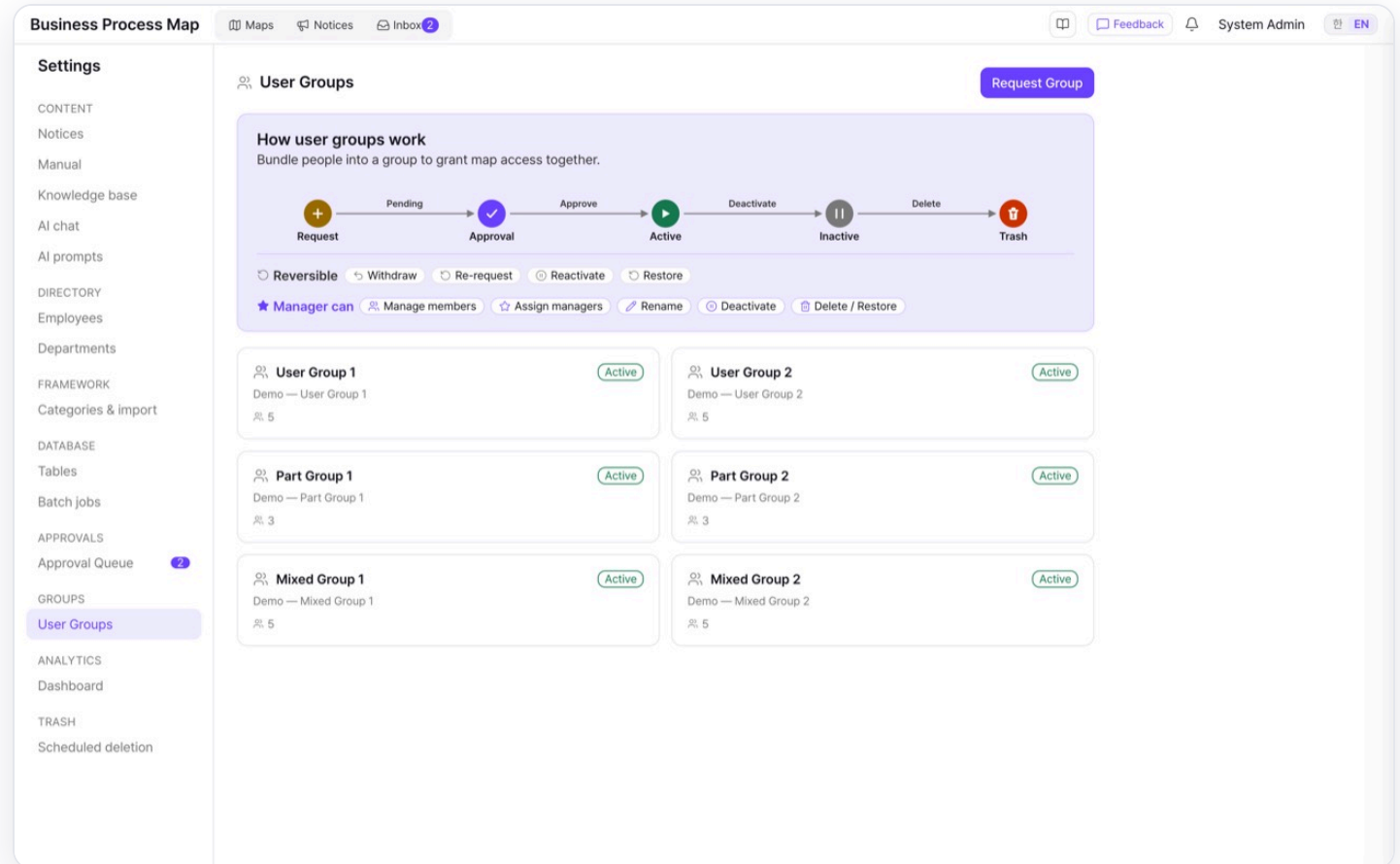
The global approval queue

- Pending requests across all maps — group creation, permission downgrades, visibility, checkout transfers, and **slot changes** (`fw_slot`) showing map · action · target · progress n/m.
- Approve/reject — this queue takes no rejection reason. Map-scoped items are also decidable by that map's approvers; this is the catch-all.
- **Slot changes are the exception** — decided by the related L5's direct admin (or a sysadmin); sysadmins decide here as the fallback.
- Renames, SP registrations and Framework **confirm requests** are handled from the Inbox — approving a confirm request confirms on the spot, so a canvas failing its gates can't be approved.



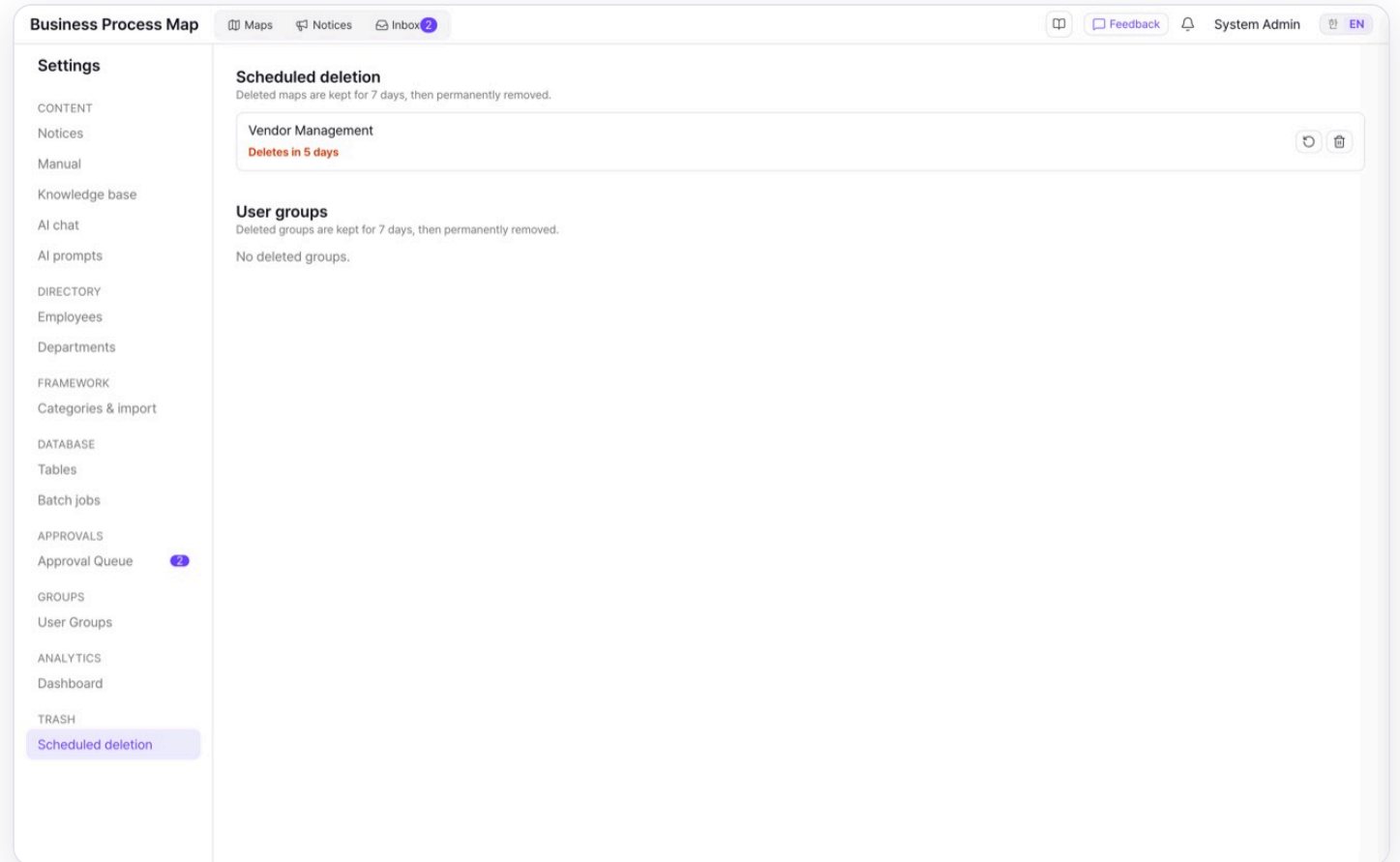
User group administration

- Group creation is request-based — **Active** after sysadmin approval.
- Check the name, members (2+), managers (1+), then decide.
- Sysadmin sees **all groups**, inactive and deleted included.
- Deleted groups purge after 7 days in the trash.



Trash & instant purge

- Deletions are soft — purged after 7 days; **Restore** brings everything back intact.
- Owners see their own; sysadmin sees **everyone's** trash.
- **Instant purge (sysadmin)** — remove immediately without the 7-day wait. Trash-state maps only; **no undo**.



Version workflow — admin powers

- Force checkout — take the lock when the holder is away (their unsaved context is lost). Checkouts never expire on their own, so this is how an absent holder's lock is reclaimed.
- As effective owner you can submit, publish, and decide checkouts anywhere.
- Withdraw is the exception — Pending/Approved only by the submitter.
- Approver reassignment — recover approver-less maps via the forced-reassign flow.
- Publishing expires the previous version (terminal) — continue via Republish.

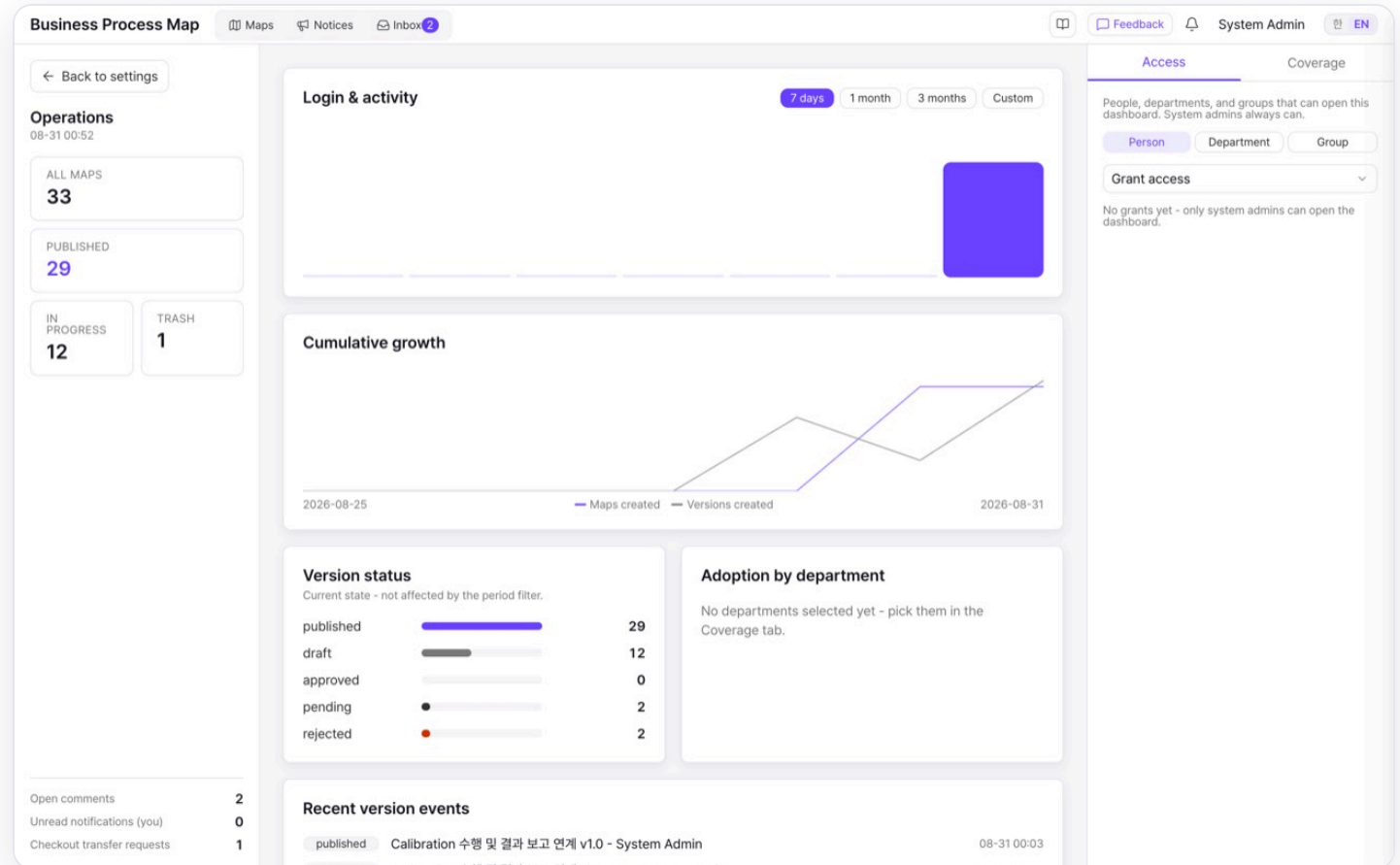
08

Dashboard

Live operational metrics and delegated access.

The operations dashboard

- Operations — maps, published, in-progress, trash, open comments, your unread notifications, transfer requests.
- Version status distribution and adoption by department (denominator picked in the Coverage sidebar).
- Login & activity, cumulative growth — 7d/1m/3m/custom period filter (snapshot metrics ignore it).
- Recent version events and AI usage (7/30-day calls, top maps).
- Access sidebar — delegate viewing to people, departments, groups (sysadmins always can).



09

Configuration Reference

backend .env and compose variables. Changes need a backend restart.

Env — auth, permissions, AI

Variable	Effect
<code>BPM_SYSADMINS</code>	Comma-separated sysadmin login IDs
<code>AUTH_MODE</code>	<code>keycloak</code> <code>ldap</code> <code>dev</code> — the frontend reads it at boot via <code>GET /api/auth/mode</code>
<code>AUTH_JWT_SECRET</code> / <code>AUTH_JWT_TTL_HOURS</code>	Idap session signing key (rotate = kill switch) · lifetime (default 8 h)
<code>DEV_ENFORCE_PERMISSIONS</code>	Enforce RBAC locally even with auth off
<code>AI_BASE_URL</code> / <code>AI_MODEL</code>	AI server (OpenAI-compatible) and model id — the in-house GPU uses just <code>glm-5.2</code>
<code>AI_MAX_TOKENS</code> / <code>AI_TIMEOUT_SECONDS</code>	Response token cap (too low = empty replies) · per-call timeout (120–180 s recommended)

Env — HR, backups, misc

Variable	Effect
<code>N8N_HR_URL</code> / <code>N8N_HR_TOKEN</code>	HR webhook URL & secret — sync activates only with both
<code>N8N_POSITION_URL</code>	EDW head-position webhook — empty disables collection
<code>HR_SYNC_INTERVAL_HOURS</code>	Auto-sync interval (default 24, 0 = off) — keep 0 for the first migration
<code>HR_SYNC_DELETE_CAP_PCT</code>	Sync deletion cap % (default 20, 0 = off)
<code>BACKUP_DIR</code> / <code>BACKUP_RETENTION_DAYS</code>	Backup path (default ./backups — point at a NAS for off-server) · retention days (default 14)
<code>MANUAL_URL</code>	Editor-toolbar manual button — hidden when empty

Env changes need a backend restart (`--reload` does not re-read .env). All deployment values live in .env — never hardcoded.

Run it steady

Check the **Batch jobs tab** each morning, **AI prompt overrides** after deploys, and **department remaps** after reorgs.